

Dhrubajyoti Ray Data Scientist +1 (412) 979-6327 dtray@tepper.cmu.edu Pittsburgh, Pennsylvania, United States linkedin github	
SUMMARY	
Data Scientist with over 3 years of experience delivering production-grade machine learning solutions in forecasting, NLP, and product analytics. Skilled in data analysis, statistical modeling, and optimization, with a proven track record of enhancing operational efficiency and decision-making through end-to-end modeling and data pipelines. Proficient in Python, SQL, and algorithm development, with expertise in deploying ML-driven systems and dashboards.	
PROFESSIONAL EXPERIENCE	
Data Scientist, Potest AI Pittsburgh, USA	Jul 2025 - Present
- Developed an inventory forecasting system for 25+ SKUs, reducing stockouts by 18% and ingredient overstock by 12% in the first evaluation cycle, aligning with metrics design & data-driven decision making - Developed a SARIMAX time-series model with seasonality and demand trends, reducing MAPE by 22% compared to baselines, demonstrating expertise in data analysis and predictive modeling.	
Data Scientist (Capstone), TIAA Financial Services Pittsburgh, USA	Jan 2025 - May 2025
- Led NLP clustering of 1.6M CFPB complaint narratives with unsupervised learning, uncovering structures that guided taxonomy redesign & enhanced regulatory reporting, demonstrating strong communication and analytical skills. - Developed a Streamlit dashboard with Python services for efficient complaint intake & automated issue tagging, achieving 30% faster resolution through ML-driven workflows, highlighting dashboard development & process optimization.	
Data Scientist 2, Micron Technology Hyderabad, India	Jul 2022 - Jul 2024
- Designed and deployed regression models using Scikit-learn and optimized Python pipelines, reducing wafer measurement time by 30% and enabling faster predictive feedback across FAB workflows, improving operational efficiency - Built end-to-end classification pipelines with XGBoost and Random Forest, improving early wafer defect detection by 15% and enhancing predictive modeling quality, aligning with data analysis & problem-solving goals. - Developed internal Streamlit dashboards integrating feature engineering, sensor patterns, and real-time wafer signals, significantly accelerating diagnostics by 2 weeks, supporting cross-functional collaboration and dashboard development. - Created scalable ETL workflows and analytics pipelines using BigQuery on GCP, enabling 40% faster processing of high-volume unstructured FAB sensor data for downstream analysis, optimizing data processing efficiency and automation. - Performed root cause analysis leveraging statistical modeling, feature engineering, & causal inference, identifying key process drivers & achieving \$10M annualized yield recovery, demonstrating strong problem-solving skills	
Product Analyst, ByteDance Mumbai, India	Mar 2020 - Aug 2020
- Developed churn prediction models using Logistic Regression & Python workflows, enabling targeted retention strategies, increasing user retention by 5% across Indian market, aligning with user engagement analytics & metrics design. - Created KPI-driven Tableau dashboards with real-time engagement metrics, supporting A/B testing and enabling faster data-driven decisions during rapid experimentation cycles, enhancing data visualization and decision-making. - Automated error-reporting pipelines using the Google Play API & Python scripts, reducing manual QA cycles by 30% and enabling efficient triaging of app crashes & performance issues, improving operational workflows and automation.	
EDUCATION	
Carnegie Mellon University, Tepper School of Business - Master of Science, Business Analytics	Aug 2024 - May 2025
Indian Institute Of Technology Kharagpur -	Jul 2017 - Jun 2022
<i>Integrated Bachelor of Technology and Master of Technology, Chemical Engineering</i>	
PROJECTS	
Full-Stack AI Knowledge Graph System	Jul 2025 - Aug 2025
- Developed an end-to-end pipeline using LLMs to extract entities and relationships from unstructured lore, then structured and stored them in a Neo4j graph database with a custom schema. - Built and deployed an interactive QA system with Streamlit and LangChain, enabling natural language to Cypher translation and real-time exploration of narrative connections.	
Customer Targeting Optimization for Bank Deposit Campaign	Mar 2025 - Apr 2025
- Designed a Logistic Regression Model with A/B testing and causal analysis to predict conversion likelihood, improving campaign precision and raising expected conversion rate from 8% to 15%. - Applied statistical inference and measurement frameworks to identify key predictors and segment high-potential customers, boosting projected ROI by \$90K through data-driven targeting.	
Peer-to-Peer Lending Risk & Return Modeling	Mar 2025 - Apr 2025
- Built Decision Tree, Logistic Regression & Lasso models on LendingClub data to predict loan defaults & reinvestment returns; improved AUC by 18% & reduced false positives by 25% by fixing class imbalance & data leakage. - Applied PCA and clustering (K-Means, DBSCAN) to segment 240K+ loans by credit behavior and risk, enabling a portfolio strategy that increased projected returns by up to 2x compared to baseline.	
SKILLS	
Programming Languages : Python, SQL, R, C++, C, HTML/CSS, Bash	
Tools and Platforms : Docker, Kubernetes, FastAPI, Flask, MLflow, Airflow, Git, VS Code, Jupyter, Streamlit, GCP, Vertex AI, AWS, Azure, BigQuery, Snowflake, Databricks, Looker, Postman, Hadoop, Spark	
Machine Learning Frameworks : Scikit-learn, TensorFlow, PyTorch, Keras, XGBoost, LightGBM, OpenCV, NLTK, SpaCy, UMAP, LangChain, Machine Learning, Statistical Modeling, Algorithm Development, Experiment Design	
Data Analysis and Visualization : Pandas, NumPy, matplotlib, seaborn, Data Analysis, Exploratory Data Analysis, Data Visualization, User Outcome Analysis, Metrics Design, User Engagement Analytics, Tableau, PowerBI	
Specialized Skills : Time Series Forecasting, Demand Forecasting, Optimization, MLOps, ETL Pipelines, Problem Solving, Cross-Functional Teamwork, Measurement Methodologies, Collaboration, Communication, Causal Inference, Privacy Compliance	